



Analysis of an Economy without Agriculture

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Abstract

The agriculture based industry is a major driver in all aspects for a country and thus proves indispensably valuable for a nation's economic growth. However, this analysis will attempt to evaluate the many differing cogs, which provide stimulating functional capability to the agricultural sector, and examine alternating circumstances, if the economic system were not supported by the agri-businesses and, or influences. As shall be repeatedly observed in the following sections, what comes out of our analysis is that recent years have seen tremendous changes, in economic development in developing countries in general, but for the agricultural sector in particular. This coincides with a renewed debate on the role of agriculture, whereby the realization of the important role of agriculture is not only reaffirmed, but also the importance of the linkages between agricultural and non-agricultural sectors, farm and non-farm activities and rural and non-rural regions is emphasized.

Introduction

An economy's vitality depends heavily on agriculture. It serves as the foundation of an economy. In addition to receiving food and raw materials, a sizable portion of the population is also given employment opportunities in agriculture. Not only that but, it also supports numerous industries that are essential to a nation's economy, including national income, international trade, and foreign exchange resources. A world without farmers or agriculture would be unimaginable for the modern world and civilization. However, if this sector is lost, nations would be fully dependent on the industrial and tertiary sectors for economic growth. We will discuss how efficient and long-lasting it will be for the entire world in this paper.

Analysis

Impacts on Food Security

Agriculture is one the world's major industries and is indispensable to food security. The world without agriculture would be a nightmare for the modern world and civilization.

The primary source of food is agriculture; it is the main thing behind the food chain. If agriculture and other agricultural activity stop, we all face the major problem of food deficiency. All the supply of fruits and vegetables starts from the hands of farmers and goes to each corner of the country. Food scarcity causes a nation's economy to suffer, and all progress is halted at one point. Fruits and vegetables are essential sources of fiber, proteins, and carbohydrates in human diets. Vitamins, such as A, C, and E, and minerals, such as magnesium, zinc, and phosphorus, are naturally occurring in many fruits and vegetables. In addition to health benefits, fruits and vegetables add flavors to the human palette. On the other hand, some fruits and vegetables are grown to provide feed for animals, from poultry to livestock. About 900 animal feed ingredients are approved by law in the U.S. (*American Industry Feed Association*). These include ingredients that come from agricultural production, including hay, straw, oils, sprouted grains, and legumes. This would be impossible without agriculture.

The produce from agriculture covers a vast portion of the food requirements, both domestically and internationally. Most countries export them, and trade for materials they do not have. If there is no agriculture, the economy may suffer from price increase, destroying the flow of trade. They will depend only on imported commodities, and when imports are not feasible, due to a pandemic or other disaster, the country risks food insecurity. For example, both major producers of staple food items, Russia and Ukraine provide 90 per cent of the wheat supply in Armenia, Azerbaijan, Eritrea, Georgia, Mongolia and Somalia. Ukraine is also a major source of wheat for the World Food Program, which provides food assistance to 115.5 million people in more than 120 countries. As the war disrupts production and export processes, these essential commodities are becoming less and less available. Similarly, at the time of the emergence of the Coronavirus pandemic (COVID-19), nearly 900 million people worldwide lack access to adequate and sufficient food. This is why the present challenge to food security is particularly serious. Within

this population lacks sufficient food, at least 155 million people suffer from severe food deficiencies, and their situation may be critically aggravated by the advance of a pandemic. In another example: India exports a good amount of agriculture products, and without agriculture, agriculture products cannot be supplied, which will impact international trade. This will have a direct effect on the nation's economy, which will be severely impacted.

Sustainable Development Goal 2 is about creating a world free of hunger by 2030 (*UN, 2015*). In 2020, between 720 million and 811 million persons worldwide were suffering from hunger, roughly 161 million more than in 2019. Also in 2020, a staggering 2.4 billion people, or above 30 percent of the world's population, were moderately or severely food-insecure, lacking regular access to adequate food. The figure increased by nearly 320 million people in just one year. Globally, 149.2 million children under 5 years of age, or 22.0 per cent, were suffering from stunting (low height for their age) in 2020, a decrease from 24.4 per cent in 2015. The number of people going hungry and suffering from food insecurity had been gradually rising between 2014 and the onset of the COVID-19 pandemic. The crisis has pushed those rising rates even higher and has also exacerbated all forms of malnutrition, particularly in children. In recent times, the war in Ukraine is further disrupting global food supply chains and creating the biggest global food crisis since the Second World War. Therefore, without agriculture, the world cannot achieve its long-term goals.

Impacts on Manufacturing and Production

For delivering inputs and outputs to one another, manufacturing and agriculture are acknowledged as the two main sectors in many types of economies. In many nations around the world, agriculture is a major contributor to economic growth, structural change, and the eradication of poverty.

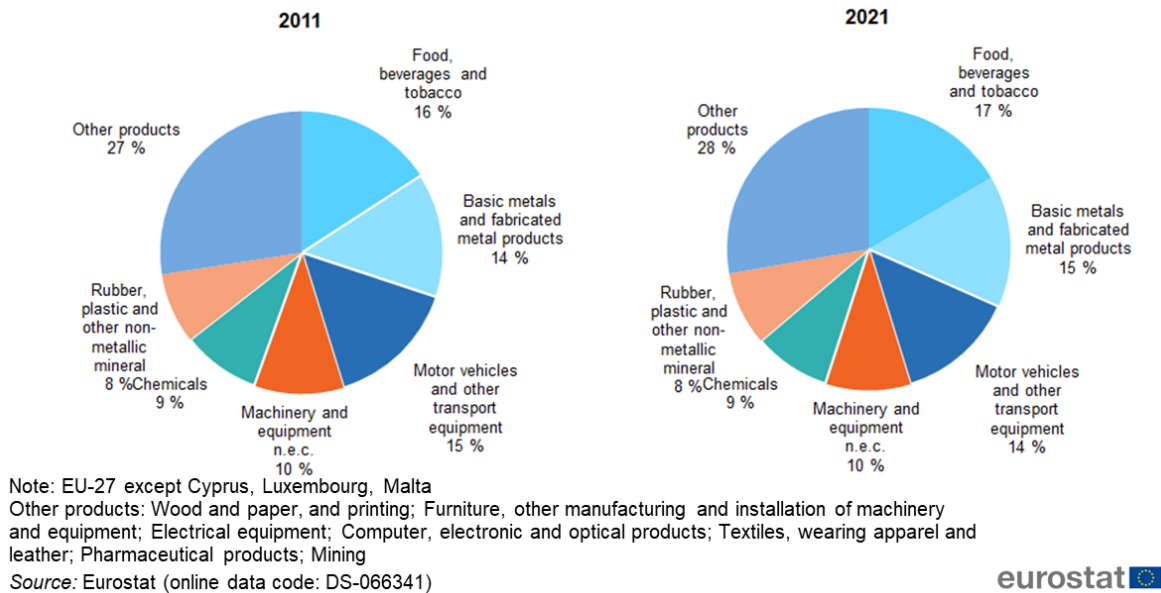
The manufacturing sector is regarded as the backbone of development in general and economic development in particular, owing to the following reasons:

- Manufacturing industries not only help to modernize agriculture, but they also help to reduce people's reliance on agricultural income by providing jobs in secondary and tertiary sectors.
- Industrial development is required for our country to be free of unemployment and poverty. This was the guiding principle behind India's public sector industries and joint ventures. It also aimed to reduce regional disparities by establishing industries in tribal and underserved areas. Exporting manufactured goods increases trade and commerce while also bringing in much-needed foreign exchange. Countries that convert their raw materials into a diverse range of higher-value finished goods are prosperous.
- Manufacturing and production plays a big role in economics. There was a time when people were mostly dependent on agriculture. But day by day new sectors of economics have been grown all over the world like the Automobile sector, pharmaceuticals sector, computer sector and so on. It is a very fast growing sector and spreading all over the world so rapidly. There are many countries that have a huge production and manufacturing based economy. Developed societies like the USA, China, and Japan have a huge market all over the world and these three countries are the three top manufacturers in 2022. Germany, Italy, France, and the UK are also doing very well in this sector and this sector is also contributing to their economy very well. Developing countries are also doing very well in this sector and this sector is contributing to their society in a very good way. For example we can mention India which is a developing country who ranked 4 in this sector all over the world. Bangladesh is also trying to develop in this sector and the development is quite impressive.

First we will talk about a developed society where manufacturing and production has a big role in their economy. We can have a look at Europe who is a big and well reputed manufacturer all over the world and it is also putting a big role in their economy. The foundation of the European economy is manufacturing. In several industrial manufacturing industries, including the production of equipment and pharmaceuticals, Europe is at the forefront. The total production makes up 83% of exports and 17.3% of the EU's GDP, making Europe the top exporter of

manufactured goods globally. Furthermore, manufacturing generates one-third of high-quality scientific publications and inspires 20% of global R&D investments.

Value of sold production by group of manufacturing activity, EU, 2011 and 2021 (% share of total sold production)



Now we will look at the pie chart shown above. Looking closely at the three major manufacturing industries inside a single nation, Greece stood out due to its 37% share of the world's total sold production in the food, beverage, and tobacco manufacturing industries in 2021. With 27% each, Croatia and Spain closely followed it.

The production of basic metals and fabricated metal products accounted for 23% of the value of Italian output that was sold. Bulgaria and Greece followed closely with 22% and 21% each, while Austria, Finland, and Slovenia accounted for the remaining 20% of the country's total value of sales production. Slovakia reported the highest share of the country's total value of sold production in the manufacturing of motor vehicles, trailers and semi-trailers, and other transport equipment, with 42%, followed by Czechia (29%), Hungary (26%), and Romania (26%). All this data indicates how manufacturing and production is impacting on Europe's economy.

Here agriculture plays a big role. During 2011 to 2021, Europe's most major manufacturing sectors were food, beverage, tobacco, rubber, chemicals and so on. All these products' raw material comes from agriculture. Therefore, if the agriculture industry did not exist in these economies, they would lose out on all the benefits gained from those productions.

Next, we will discuss a developing society whose economy is likewise influenced by industry and production. We may examine South Asia's economy, which is attempting to have a positive effect on its economy, through the secondary sector.

In South Asia manufacturing output for 2018 was \$527.02 billion, a 3.07% increase from 2017. For 2019 it was \$513.31Billion, a 2.6% decline from 2018. For 2020 it was \$492.69Billion, a 4.02% decline from 2019. For 2021 it was \$595.42B, a 20.85% increase from 2020. This indicates that this sector is growing in this region day by day and putting a good impact on its economy. Many big companies are investing here for its lower labour costs, enhanced manufacturing capabilities and diversification.

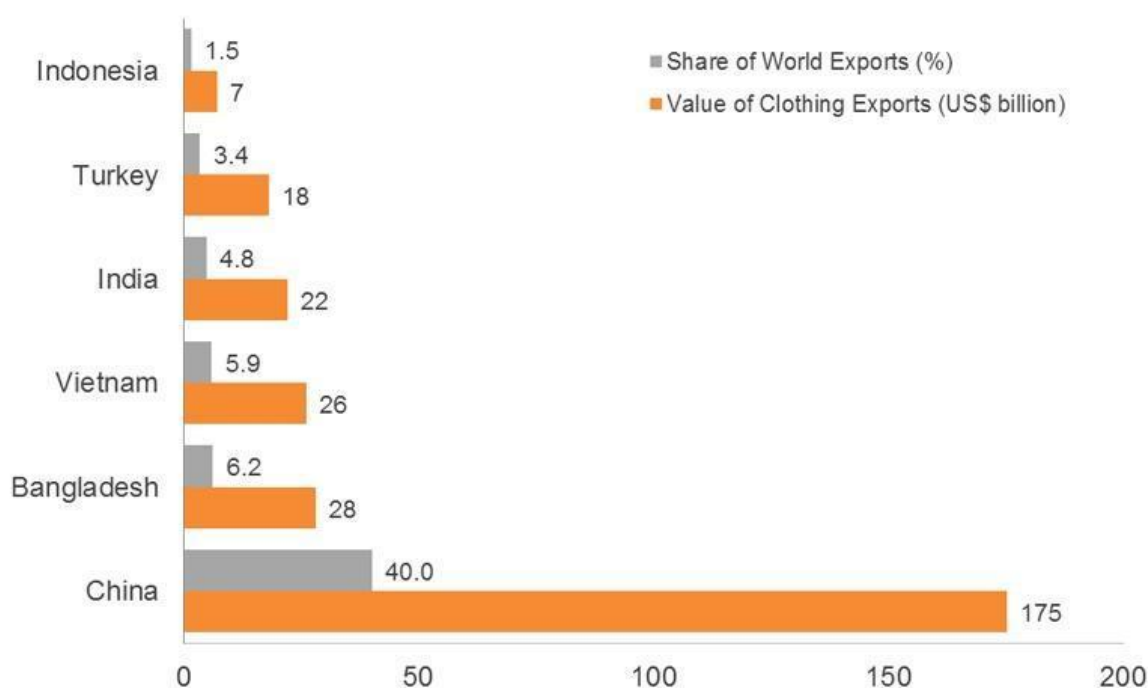
In India, manufacturing's share of GDP has remained constant over the last two decades at 28% of GDP, out of a total of 28% for the industry, which includes 10% for mining, quarrying, power, and gas. Various initiatives have been undertaken by the government for the manufacturing sector in India. In Pakistan, the industrial sector is dominated by manufacturing, which accounts for 12.4% of GDP. In Bangladesh the manufacturing sector currently accounts for 17% of the value added to the country's GDP and 90% of exports. Over the past six years, manufacturing has consistently accounted for around 18% of Sri Lanka's GDP. Clothing, rubber and plastic products, and fabricated metal products have been the fastest-growing segments.

Next we will talk about the impacts of the Manufacturing and Production sector on the economy of Bangladesh. 90% of exports and 17% of the value added to the nation's GDP are currently contributed by the manufacturing sector. This industry accounts for roughly 15% of employment in the nation, has significantly increased the participation of women in the labour force, and is creating job prospects in an industry where a sizable portion of workers are still self-employed.

Indeed, nearly half of urban working women under the age of 25 are employed in the manufacturing sector, which generates significant multiplier effects by creating approximately two direct and induced jobs for every direct job created.

Since the early 2000s, Bangladesh has evolved into the world's second largest clothing exporter after China, accounting for about 6% of global clothing exports. An important sector of Bangladesh's economy is the "readymade garments" (RMG) industry, which accounts for over 80% of all export revenue.

World's Top Clothing Producers in 2015



Source: World Trade Statistical Review 2016, World Trade Organisation

The graph shown above indicates the RMG industry's situation all over the world where Bangladesh was above Vietnam, India, Turkey and Indonesia. Agriculture is also its source of raw material.

Based on Bangladesh export statistics, here are the top most exported products of Bangladesh.

Product	Total Export in %
Readymade Garments	83.4
Jute and Jute Manufacture	3.6
Fish, Shrimps, and Prawns	1.4
Leather and Leather Manufactures	1.8
Home Textile	1.7
Vegetable Products	0.7
Pharmaceutical Products	0.4
Plastic and Plastic Products	0.3
Bicycle	0.3
Terry Towel	0.1

This chart indicates that the raw materials of most of the items manufactured by Bangladesh are produced from agriculture. This is because clothes today are made from a wide range of different materials. Traditional materials such as cotton, linen and leather are still sourced from plants and animals, which have to be carefully cultivated and, or farmed.

Bangladesh manufacturing output for 2018 was \$66.85B, a 13.37% increase from 2017. In 2019, it was \$74.49B, an 11.42% increase from 2018. In 2020 it was about \$77.02B, a 3.39% increase from 2019. In 2021 it was \$88.40B, a 14.77% increase from 2020. In 1972 the breakdown of the industrial sector value added to GDP in Bangladesh was 6.6%, where in 2018 it went to 19.9%. These indicate how this sector is putting an impact on the Bangladesh economy. Without agriculture, the raw material of most items is mostly impossible to find. It gives a clear view of how agriculture is contributing to this sector.

Agriculture and manufacturing are recognized as the core sectors in diverse sorts of economies. One is dependent on another. Consequently, it can be denoted that maintaining both sectors effectively is crucial for a country's economy to function efficiently.

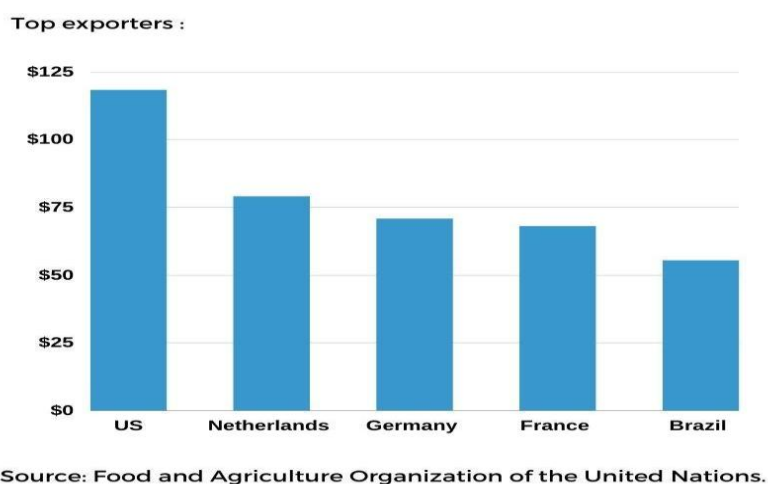
Impacts on the Balance of Payment

First of all, increased production levels are essential for the security and well-being of a nation's people because agriculture is one of the biggest and most important sectors in the world. Besides, it is well established that exports and imports are significant because they collectively make-up a nation's trade balance which has an effect on the state of an economy, as a whole. Both imports and exports expand steadily when the economy is strong. These often denote a resilient and powerful economy. Agricultural products cover the following commodity categories: food and live animals, meat and meat preparations, dairy products and birds' eggs, fish, crustaceans, mollusks, and preparations thereof, cereals and cereal preparations, vegetables and fruits, sugar, sugar preparations and honey coffee, tea, cocoa, spices, and manufactures thereof, Feedstuff for animals (excluding un-milled cereals). And thus, the production of all these commodities significantly adds to the output of a nation.

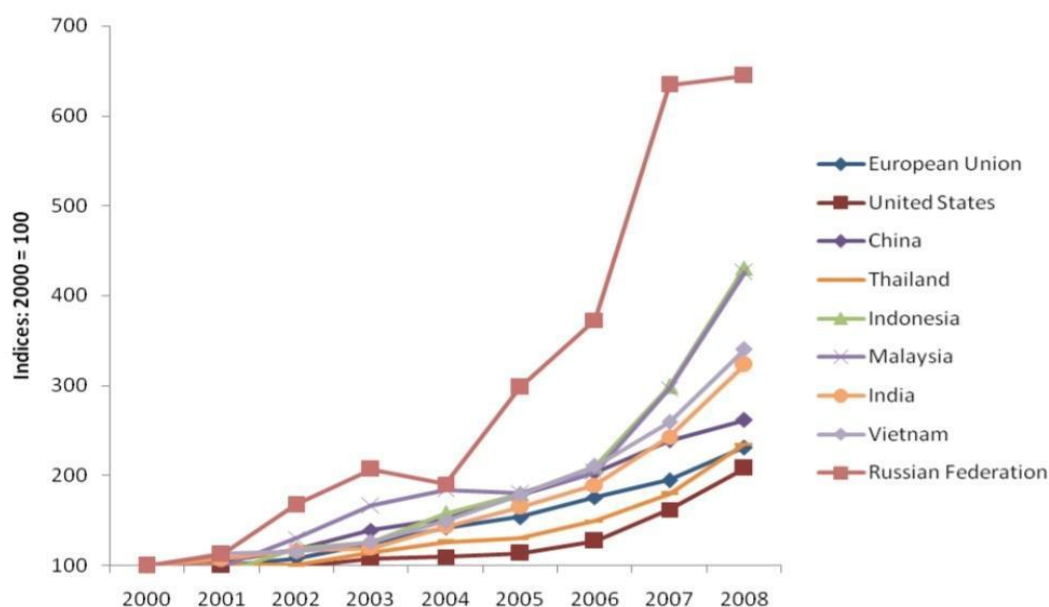
The following section will explore several examples of the category of agricultural commodity that is being produced in different regions of the world and will showcase the contribution and dependence; they exude to individual and the global economy – in the process illustrating the importance of exports and imports of agricultural products.

The top producers of the major cereal and vegetable crops are the US, China, India, and Russia. With \$118.3 billion in exports as of 2019, the United States is the world's leading exporter of agricultural products. The sheer tonnage produced and the financial value of the commodities produced is two of the many metrics used to evaluate agricultural output. It's crucial to consider both since frequently, crops with high cash values do not correspond to commodities that are essential to the food supply of less developed nations. China, India, the United States, and Russia

frequently rank as the leading producers of the major cereal and vegetable crops. The fact that China is the world's top producer of rice, as well as the top producer of wheat and the second-largest producer of corn, as well as the top producer of numerous vegetables, including onions and cabbage, is surely astounding. The United States ranks first in corn, first in soybeans, and second in wheat production overall. But there are other crucial crops. Similarly, crops like barley, rye, and beans and pulses are not as essential in the United States as they are in nations like Russia, Germany, and India. Most of Africa and Asia grow millet as a significant crop, with India and Nigeria producing the most of it.



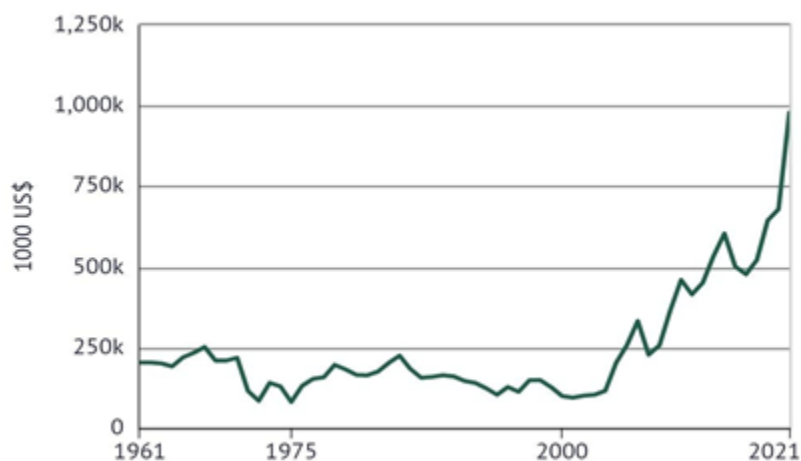
Moreover, the majority of the region's agricultural exports have only been distributed among a small group of Asia-Pacific nations. China was the leading exporter in 2009, accounting for 87% of all exports from the Asia-Pacific region. The majority of the shares were held by Southeast Asian nations, followed by Australia, New Zealand, and East and Northeast Asian nations. With regard to growth trends, Asia-Pacific agricultural exports have kept pace with those of the rest of the world. Indicators show that between 2000 and 2008, the top 50 global agro-exporting countries grew at an average annual rate of 19%, while the top 10 Asia-Pacific nations grew at an average annual rate of 18%. One of the top ten countries to this high export growth is India, followed by Indonesia, Malaysia, the Russian Federation, Thailand, and Vietnam.



Source: COMTRADE 2011

(Agricultural export Indices for Selected Countries/Regions 2000-2008)

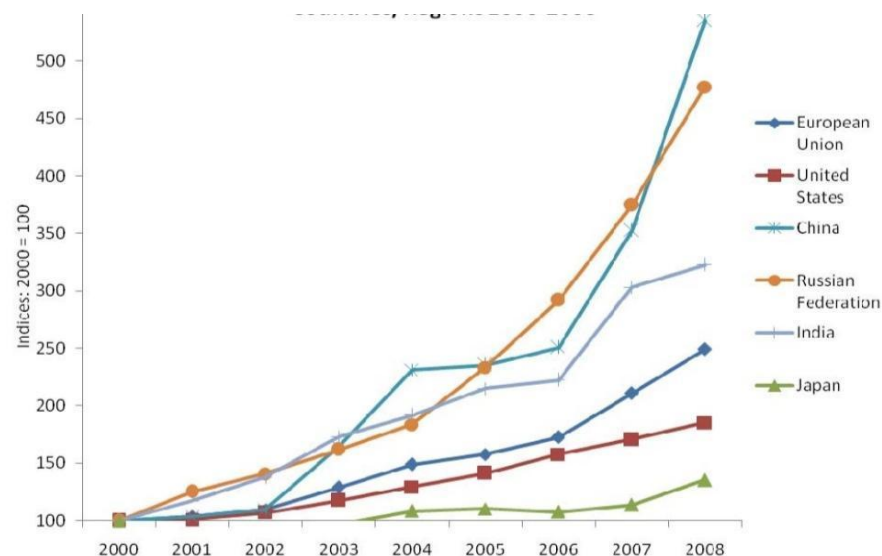
On the other hand, Bangladesh's agricultural exports climbed from 84,138,000 US dollars in 1972 to 97,931,336,000 US dollars in 2021, rising at an average yearly rate of 7.50%. Bangladesh's top export products are Ready-made Garments (83.4%), Jute and Jute Manufacture (3.6%), Leather and Leather Manufactures (1.8%), Fish, Shrimps, and Prawns (1.4%).



The Export of agricultural products from Bangladesh (1961-2021)

In light of this, if the agriculture sector were to be excluded from the economy (that is, if there is no agricultural export then the GDP of a country will decrease). Also, if countries do not export agricultural goods and services, they will not have a competitive or comparative advantage.

On the other hand, agricultural import means to bring in agricultural products, goods, etc. from another country for sale or use. From our analysis: China, the United States, and Germany were the top three international importers of agricultural products in 2020. 193.5 billion dollars' worth of imports came from China. German imports totaled less than 100 billion dollars, while American imports were less than 150 billion. Moreover, in some Asia-Pacific nations compared to others, agricultural imports have increased more quickly. For instance, between 2000 and 2008, the average annual agricultural imports into Japan and China respectively placed them fourth and thirteenth globally. According to indices related to import value, agricultural product imports in the top 10 Asia-Pacific countries increased at an average annual rate of 12% from 2000 to 2008, while the comparable figure for the top 50 importers of the world is 13%. It is important to note the growth of the three fastest growing Asia-Pacific countries in terms of agricultural imports: China, Russian Federation and India.



Source: COMTRADE 2011

(Agricultural Import Indices for Selected Countries/Regions 2000-2008)

Besides, food-grains (rice and wheat), edible oil and oilseeds, raw cotton, milk and milk products, spices, sugar and coconut oil are the main agricultural imports of Bangladesh. The top five imports into Bangladesh are refined petroleum (\$2.64 billion), raw cotton (\$1.8 billion), wheat (\$1.28 billion), light rubberized knitted fabric (\$1.09 billion), and heavy pure woven cotton (\$975 billion), with the majority of these goods coming from China (\$15 billion), India (\$7.91 billion), Singapore (\$2.2 billion), the United States (\$1.79 billion), and Indonesia (\$1.69 billion). As well as, the largest importer of heavy pure woven cotton (\$975M), heavy mixed woven cotton (\$692M), light pure woven cotton (\$615M), non-retail synthetic staple fibers yarn (\$306M), and woven fabric made of synthetic staple fibers (\$194M) in 2022 Bangladesh.

So, if the agricultural sector were to be eliminated from the economy and if there were no agricultural imports then it would become a matter of concern for a country. Imports are crucial to the economy because they enable a nation to provide its market with goods from other nations that would otherwise be unavailable, rare, expensive, or of low quality.

In consequence, the second Sustainable Development Goal, to eliminate hunger by 2030, places a great deal of importance on agricultural commerce. Trade can actually play a significant part in accomplishing this crucial goal, despite the fact that it is generally perceived as posing risks to it. Trade benefits nations in a variety of ways by enabling them to benefit from their radically different factor endowments. Land-rich nations can export, while land-poor nations can benefit from imports that are produced much more effectively. Without agriculture trade these would not happen.

Impacts on Employment

One of the major industries in the world in terms of employment is agriculture. There are many employment opportunities in farming, including those for farmers, technicians for farming equipment, common workers, scientists, and many others. 1.3 billion People worldwide, 97% of them in developing nations, are employed in agriculture. Depending on the region, farming accounts for anywhere between 30 and 50 percent of household income in rural areas. (FAO)

Particularly where agricultural production goes beyond subsistence and provides the food market or businesses that process primary products, there is a significant employment incentive for regional economic development. Agriculture is inherently connected to employment in industries that are closely related to farming, such as retail commerce, services (transport, banking), and food processing. Agriculture occupations are prevalent in most sectors and reduce the unemployment rate in both developing and industrialized nations. CMIE data from the consumer pyramid household survey shows the share of agriculture in total employment has gone up from 35.3% in 2017-18 to 36.1% in 2018-19 and then to 38% in 2019-20. (*The Economic Times*)

In the United States, the economy can be significantly impacted by agriculture as the agricultural and food industries supported about 20 million full- and part-time jobs in 2020, or 10% of all occupations in the country. 19.7 million full- and part-time jobs, or about 10.3% of all employment, are supported by agriculture and allied industries in the United States (*U.S. Department of Agriculture's Economic Research Service*). Additionally, according to the USDA, crop-related cash receipts for 2020 will amount to almost \$198 billion. The food and agriculture industry is a crucial driver of economic activity due to its connection with other industries, such as transportation, energy, and chemical (*CISA*). According to the USDA, food services and eating and drinking establishments accounted for: the direct and indirect contributions that agriculture makes to the overall U.S. gross domestic product (GDP), as well as 10.5 million jobs in 2020, the largest share among all categories within the agriculture and food sectors. Agricultural products are utilized in the production of a wide range of goods, including textiles, personal care and cleaning products, construction materials, fuels, and more. About 1.7 million people are employed by food and beverage manufacturing businesses in the United States. Farms and related industries contributed more than \$1.1 trillion to GDP, a little over 5% of the GDP, in 2019 (*USDA*). Termination of the agriculture sector will hamper all the employment rates generating a large number of unemployment rates as well as stagnation of many sectors that are indirectly related and dependent on agriculture.

Agriculture contributes significantly to job growth all around the world. According to a study on the significance of agribusiness from BBVA, a corporate and investment bank, agriculture accounts for 25% of exports in developing countries in Latin America, or around 5% of their

regional GDP. Jobs and economic activity are generated by this activity in these nations. In the least developed nations, 66 percent of all economically active people were employed in agriculture in 2009, more than double the percentage of GDP from agriculture. A given rate of GDP growth attributable to agricultural growth decreases poverty and creates jobs in resource-poor low-income countries (apart from sub-Saharan Africa) five times more than a similar dose of GDP growth due to non-agricultural growth. Agricultural growth is 11 times more efficient in sub-Saharan Africa. Increasing agricultural productivity and production is therefore still essential for eradicating poverty in low-income nations, in particular (*SOFI*). Therefore, there is an immense possibility of an increase in poverty, as well as, unemployment rates in these types of countries, if we try to establish an economy without agriculture.

In economies focused on agriculture, only 20 to 30 percent of rural income comes from non-agricultural sources, but in countries that are urbanizing, that percentage can reach 60 or 70 percent. The majority of households engage in rural non-farm income producing activities, according to examination of RIGA (*Rural Income Generating Activities*) data; and usually between 50 and 60 percent in Asia and Latin America, and 25 to 50 percent in sub-Saharan Africa. However, in Asia and Latin America, only 20–25 percent of rural households and 10–20 percent in sub-Saharan Africa get more than 75 percent of their income from non-farm sources. Therefore, engagement in the non-farm sector for the majority of households is either seasonal or part-time and serves to reduce risk and diversify sources of income. In general, rural households have one foot in the agricultural sector and the other in the non-agricultural economy.

The rural non-farm sector has traditionally risen quickly in areas where agriculture has had strong growth. According to the research, each dollar of extra value added in agriculture results in second-round income benefits of between 30 and 80 cents elsewhere in the economy, (depending on elements like population densities and the availability of excess labor). The relationship between agriculture and other sectors changes as a country develops. At low levels of development, it promotes growth in other economic sectors. As a country develops, the relationship becomes more beneficial to both parties. Eventually, agriculture loses its significance as a driver of economic growth. On the other hand, slow agricultural revenue growth results in low consumer demand, constrained demand for agricultural inputs, constrained

expansion of agro-processing and stagnant wages. In these conditions, little economic dynamism can be anticipated, and poor rural households will be forced to resort to survival techniques including relocation and low-return non-farm activities. All of this shows that there may be potential to promote the expansion of the non-farm economy and establish a positive cycle of rural development and job creation, especially in nations with a large agricultural sector.

Agriculture is a subsistence activity in many underdeveloped nations, including the Nigerian economy. The majority of people who live in rural areas rely on revenue from agricultural production to survive. The agricultural industry employs around 70% of all workers, hence agricultural reform is very important for lowering unemployment and promoting national development. Every rise in income or per capita agricultural output always raises the incomes of the poor and lowers the number of people in this region subsisting on less than \$1 per day, which increases capital creation.

Agriculture expands the range of revenue sources available to different demographic segments by promoting the development of industries and expanding employment opportunities. A rise in income triggers capital formation, which fuels the expansion of the non-agricultural sector. Through the export of agricultural products, which speeds up the development of other economic sectors, it contributes to foreign exchange revenues. The industrial growth, a byproduct of capital formation, serves as a catalyst for the creation of new industries and an increase in employment opportunities for people in both the agricultural and non-agricultural sectors. This situation promotes the growth of related industries like the banking, insurance, and service sectors. Overall, all of them will have a multiplier impact in order to increase integration and links between the various economic sectors.

Now, speaking in terms of Bangladesh, for a person's livelihood, employment, and GDP contribution, agriculture is essential. Over the past ten years, it has made less of a contribution, from 16% in 2010 to 16% in 2020. The sector has however remained resilient in terms of profitability and productivity despite the ever-increasing population, which is expected to increase from 147.6 million in 2010 to 164.7 million in 2020 when the impact of the pandemic and climate change are also taken into account. All of these agricultural sub-disciplines began to

take shape as commercial farms run by young, forward-thinking women entrepreneurs, not merely side projects. The GDP decreased in relation to the primary agricultural sector, but farming as a whole—including its subsectors, service, and industrial sectors—has grown significantly in importance. Over the past 20 years, the value of agricultural produce has climbed at a rate of 3.54 percent annually (1999-2019). For 40% of the labor force, the sector continues to be the primary source of employment. Therefore, in terms of employment, agriculture continues to be the largest sector. The Labor Force Survey is carried out by the Bangladesh Bureau of Statistics (BBS) to evaluate the general employment condition (LFS). The number of people over 15 who are economically engaged is 5.67 crore, according to the most recent BBS poll, the "Labor Force Survey 2010". Out of this, 5.67 crore individuals (3.79 billion men and 1.62 billion women) are employed in various occupations, with agriculture still accounting for the biggest proportion, 47.33 percent. The share of the employed labor force above 15 years by different sectors of the economy according to the Labor Force Surveys 1995-96, 1999-00, 2002-03, 2005-06 and 2010 is shown in Table below.

Table 7: Share of Employed Labor Force above 15 Years by Sectors

Sector	1995-96	1999-00	2002-03	2005-06	2010
Agriculture, forestry and fishery	48.85	50.77	51.69	48.10	47.33
Mining and quarrying	-	0.51	0.23	0.21	0.18
Manufacturing	10.06	9.49	9.71	10.97	12.34
Power, Gas and water	0.29	0.26	0.23	0.21	0.18
Construction	2.87	2.82	3.39	3.16	4.79
Trade, hotel & restaurant	17.24	15.64	15.34	16.45	15.47
Transport, maintenance & communication	6.32	6.41	6.77	8.44	7.37
Finance, business and service	0.57	1.03	0.68	1.48	1.84
Commodities and personal services	13.79	13.08	5.64	5.49	6.26
Public administration and defense	-	-	6.32	5.49	4.24
Total	100.00	100.00	100.00	100.00	100.00

Source: BBS Labor Force Survey (LFS)

However, the selling and processing of agricultural products account for the majority of the increase in the service sector. This is mostly caused by the agricultural sector's increased commercialization and diversification. As a result, an increase in agricultural income of around 10% causes an increase in non-agricultural income of approximately 6%. This is a major distinction as it indicates that Bangladesh's economic growth is being fueled by agriculture. Consequently, if the agricultural sector is eliminated it would put an abrupt end to this gradual stimulation. Furthermore, agriculture occupations in emerging nations contribute to lowering

high unemployment rates. Evidence suggests that concentrating on agriculture is much more beneficial than spending in other sectors when it comes to decreasing poverty (*First Seed Foundation*). If agriculture is eliminated then all of these sectors will face a negative impact resulting in the decrease of economic growth.

Impacts on Environment

Agriculture plays an essential role in our lives. Agriculture is not only related to farming but also forestry, dairy, beekeeping, fruit, mushroom, fisheries, and livestock. Agriculture provides most of the world's food and raw materials. Furthermore, it also provides employment in many countries. Most of the underdeveloped and developing countries' economies mainly depend on agriculture. However, aside from these beneficial aspects, agriculture is also the cause of a number of environmental issues. Population growth throughout the world increases food demand which creates additional pressure on food production. Agricultural capacity is not enough to meet the food demand of the entire world. So Pesticides, fertilizers and other chemicals are used rapidly to increase food production. Additionally, the negative consequences of agriculture on the environment are: water pollution, climate change, loss of biodiversity, soil degradation, increased pollution and waste etc.

One of the negative consequences of agriculture on the environment is land and soil degradation. Land erosion is the main cause for land degradation. Modern industrial agriculture practices, increase deforestation, overuse of land, unsustainable farming causes extreme land erosion. Half of the top fertile lands are lost in the last 150 year and globally 25% land is already degraded. About 24 billion tons of fertile land is degraded every year. If this process continues then 95% of the world's land will degrade within 2050. In Europe 80% of land degraded due to agriculture and approximately 40% of land is already degraded. For an increase in agricultural production Africa lost 5 million hectare of forest and 230 million hectare land is degraded every year. Around Africa, Niger has had the most degraded land in the world where more than 75% land is highly degraded. Similarly, due to degradation of land Asia lost US\$84 billion. However, India, China, Iran, Mongolia, and Pakistan are the most land degraded countries in Asia. In India 30% of land or 97 million hectares is already degraded. Bangladesh has also seen severe land degradation as a

result of rising agricultural activity. An estimated 6 million hectares, or 43% of the country's land area, have been damaged. Land degradation has severely hampered the environment and ecosystem. However, if agriculture does not lead for land degradation and land erosion then many environmental issues such as loss of soil biodiversity, soil infertility, soil salinization, creating new deserts, floods will decline and environmental sustainability. (*Source: croplife, FAO, UN*)

Deforestation is one of the harmful effects of agriculture on the environment. Nearly 90% of global deforestation is caused by agriculture. However, 48% and 32% of deforestation are caused by commercial and subsistence farming, respectively. The increase of agricultural land is to blame for more than half of the global loss of forests. In the regions of Asia and Africa, this rate is greater. Cropland expansion is to blame for more than 75% of the loss of forests. Three-fourths of deforestation in South America is caused by the raising of cattle. Seventy percent of all animal and plant species find refuge in forests. Deforestation causes the extinction of over 50,000 plant and animal species annually. Deforestation destroys the natural habitat of species ecosystems by causing the extinction of plants and animals. Deforestation is also to blame for the disruption of the water cycle. 30% of fresh water comes from tropical rainforests. However, groundwater levels and humidity are declining as a result of deforestation. Deforestation contributes to an increase in greenhouse gas emissions. Deforestation contributes to an annual 11% worldwide increase in greenhouse gas emissions. Floods and desertification are two additional effects of deforestation. Considering all of the listed issues, a complete reduction in agricultural practices will reduce greenhouse gas emissions, safeguard millions of plant and animal species, promote water cycling, and maintain ecological function if they are unable to cause deforestation. (*source: Wikipedia, BYJU`s*)

Water contamination is another adverse impact of agriculture on the environment. Globally, 70% of all freshwater is used for agriculture. A variety of agricultural practices can contaminate water. Groundwater and surface water contamination is a result of agricultural operations. Farmers employ an excessive amount of pesticides, chemical fertilizers, nutrients, organic carbon, metals, and diseases that contaminate groundwater and flow into surface water bodies, in order to boost agricultural productivity. In addition, agricultural pollutants such crop leftovers,

leaf litter, forest debris, and animal manure severely contaminate water. Agriculture frequently pollutes water in Asia, Africa, and South America. In Asia, rivers contain 50% of the region's total nutrients, compared to 25% for a moderate level. Groundwater and surface water in Bangladesh are severely contaminated by various agricultural practices. Additionally, the ecology is impacted by water contamination. As a result, many nations are seeing a loss in aquatic biodiversity. Despite the fact that polluted rivers have low oxygen levels, they are more susceptible to the negative impacts of climate change. As a result, if the agriculture industry ceases water pollution, millions of aquatic species, plants, and the water ecosystem will be saved. (Reference: World Water Forum)

Air pollution is also a result of agriculture. Ammonia is the primary cause of agricultural air pollution. It enters the environment through cattle feces and fertilized crops. However, 40% of world emissions come from cattle. A total of 6 million tonnes of greenhouse gas are produced by cattle farming. Additionally, agricultural waste is bad for the air. Farmers, who gather agricultural rice straw, often burn the waste to get rid of it, which significantly increases carbon monoxide production. Several resources are required for the use of agricultural mechanization, including smoke-producing fuels like diesel and gasoline, as well as electricity. Agriculture presently accounts for 11% of US emissions. The area of Asia also produces the most greenhouse gas emissions. Bangladesh's greenhouse gas emissions in 2019 were 21,594,000 m. tons, up 3.82% from the year before. Agriculture will exert less strain on the environment if it does not contribute to air pollution. Additionally, it will reduce greenhouse gas emissions and aid in the fight against climate change. (Reference: FAO)

Agriculture and Economic Development

According to the culmination of the analysis so far, agriculture plays a consequential role, especially for developing economies, in terms of contributing to the achievement of the macro economic objectives; namely – providing a source for employment, stabilizing commodity prices, acting as a source of income in the form of exports, and distribution of income – all of which adds to the GDP of a country and stimulates the economic growth, as has been repeatedly demonstrated.

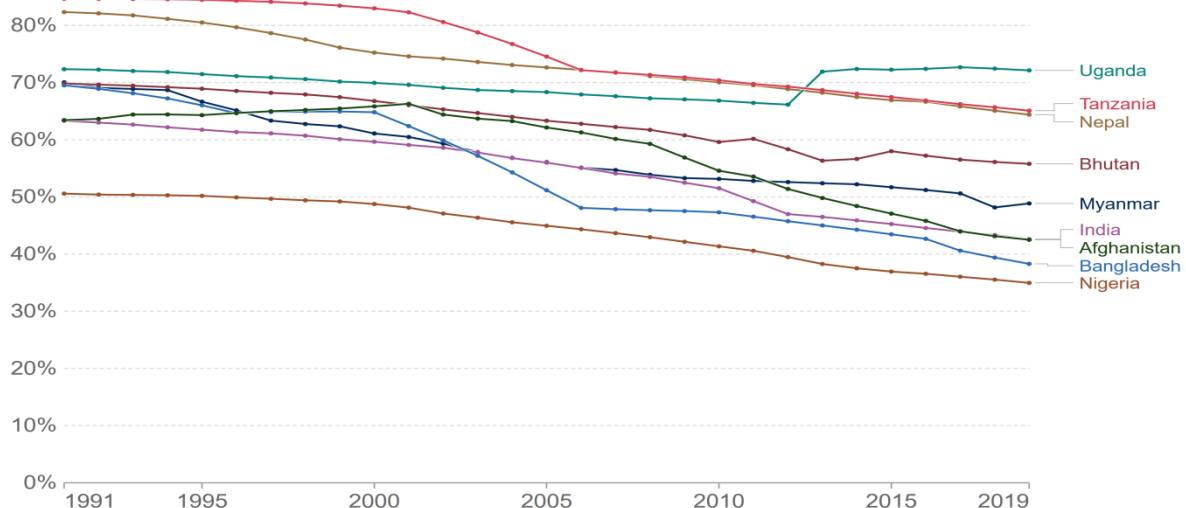
All nations, whether underdeveloped, developing, or even developed, rely on agriculture as their primary means of supplying food. The demand for food is rising quickly as a result of the strain that undeveloped and emerging nations' populations are under and their rapid growth. The economy's growth rate is shown to be negatively impacted if agriculture is unable to keep up with the growing demand for food goods. Therefore, increasing food supply through the agricultural sector is crucial for a nation's economic development.

Furthermore, the primary sector is a major source of employment for developing nations. The production of agricultural commodities provides a livelihood to millions of farmers and their families. In poor and emerging nations, agriculture offers rural residents a significant number of job opportunities. Therefore, it can be deemed as a crucial source of income. The importance of agriculture for developing economies can be further exemplified in the figure below.

Share of the labor force employed in agriculture

Agriculture includes the cultivation of crops and livestock production, as well as forestry, hunting, and fishing. Employment includes anyone engaged in any activity to produce goods or services for pay or profit.

Our World in Data



Source: Our World in Data based on International Labor Organization (via the World Bank) and historical sources
OurWorldInData.org/employment-in-agriculture • CC BY

From the data, it can be extrapolated that in recent years agriculture may actually amount for over 70% of the employment source in some economies, such as, in Uganda. Even-though, there is a general trend of reduction in the proportion of employment in the primary sector, it still holds an impactful cadence on the overall labor force: where the lowest percentage of

employment can be seen in Nigeria and Bangladesh which is still just under 40%. As a consequence, if the agricultural sector were to be opted out of the economy, these nations would lose a staggering amount of employment, lowering living standards, decreasing output and eventually lowering the GDP and economic growth of the country.

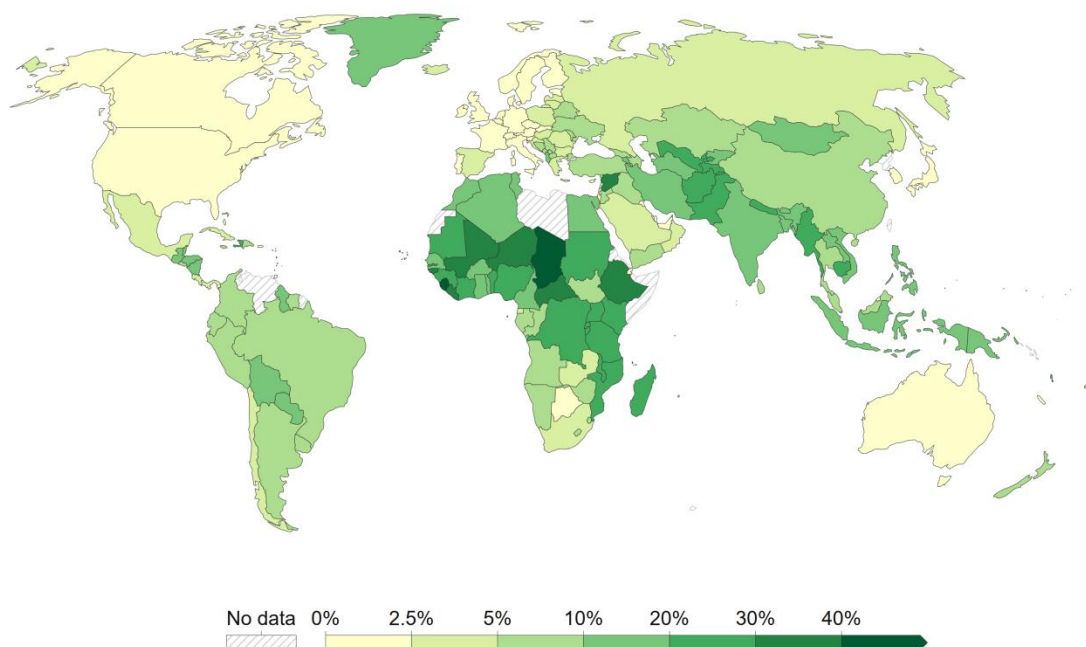
Improvements in agriculture are required to increase the availability of raw materials for agro-based businesses, particularly in emerging nations. The industrial output is impacted by the lack of agricultural products, which raises overall prices; as a result it will prevent the economy of the nation from expanding.

Additionally, agricultural goods form the foundation of a large number of companies, including flour mills, rice dealers, oil, bread, meat, milk products, sugar manufacturers, jute mills, and textile mills. Thus, if the primary sector did not exist, neither would these firms or the employment, investment or injection they make to the circular flow of income, in an economy.

There is a larger economic disparity between rural and urban sections of a country that is mostly agricultural and overpopulated. Giving agriculture more importance is vital to lessen this economic inequality. The majority of rural residents' incomes would rise as a result of agriculture's success, potentially reducing the income gap to some extent. The rise of the agricultural sector will likely improve the buying power of farmers, which will aid in the expansion of the nation's non-agricultural sector. The market for industrial goods will expand as a result of agricultural improvement. A rise in agricultural productivity increases the income of the rural population, which in turn increases demand for industrial goods and fosters the growth of the industrial sector. A market will be created for increasing manufacturing. It is common knowledge that the majority of people in developing nations depend on agriculture, and it is up to them to be able to afford to buy the items produced. As a result, it will aid in promoting the expansion of the non-agricultural sector. Similar to this, an increase in cash crop yields may open the door to the development of an exchange economy, which might aid in the expansion of the non-agricultural sector.

Share of GDP from agriculture, 2020

This is measured as the value added from agriculture, forestry and fishing products as a share of gross domestic product (GDP).



Source: World Bank and OECD

OurWorldInData.org/agricultural-production • CC BY

Lastly, from the map above, it can be observed that agriculture in the African region has been extensively important; as almost half of their GDP is extracted through the operation of this sector. As a result, opting agriculture out of the economy can be deemed a practical impossibility for these nations. Not only that, but nations, which are heavily invested in agriculture, have already been cultivating that sector for many decades, subsequently developing a comparative advantage on the world stage. For instance, Cote d'Ivoire has a major comparative advantage on the production of cocoa and thus, is the biggest supplier of cocoa in the world (*WorldAtlas, 2022*). Logically, losing the primary sector would not only affect Cote d'Ivoire, but also the global market for all cocoa related commodities.

Poverty Alleviation

The role of agriculture has been restricted in contemporary anti-poverty measures adopted by nations and development agencies, partly due to a failure to acknowledge the significance of the connections between agricultural development, food availability, dietary patterns by the poor, and poverty reduction. Because the poverty line is frequently expressed in terms of calories, reducing poverty is defined in part by this aspect. However, increasing the caloric intake of the underprivileged has a favorable impact on their health, productivity, and investment in human capital. Empirical evidence illustrates this importance, but a more general case can also be made.

The case builds on three empirical relationships: between agricultural growth and poverty alleviation; between increases in domestic food production and improvements in nutrient intake; and between agricultural productivity and productivity growth in the rest of the economy. It has long been established that, for a given level of income per capita, a higher share of GDP originating in agriculture contributes to a more equal distribution of income, and recent empirical work confirms that message. If aggregate growth is not sacrificed, an agriculture-driven development approach increases income distribution to the poor, making it more pro-poor. The first step in ending the cycle of poverty is to implement such a plan.

However, despite substantial historical research on the subject, the role of agriculture in economic growth is complex and ambiguous. The structural transformation as a general equilibrium process is difficult to comprehend from the perspective of the agriculture sector, which contributes to the disagreement. The diversity of agricultural resources and the wildly dissimilar cropping patterns seen in Latin America, Africa, and Asia further contribute to debate regarding the contribution of agriculture to economic development (not to mention the diversity within these regions). Expecting much of a common role in such a variety of environments is impractical. It is simple to understand why there is so little agreement in academics, given the stark contrasts in global developmental phases and, therefore, the significantly diverse roles that agriculture plays in countries at various degrees of economic maturity.

Conclusion

Conclusively, the amalgamation of the entire study leads to the conspicuous assessment of agriculture's traditional functions as a source of food, employment, export revenue, savings and investment capital, as well as the production of basic commodities for developing industries, have long received attention from policymakers.

However, the impact of agriculture frequently goes beyond these immediate, market-mediated effects. Indirect non-commodity contributions from agriculture, such as public goods, social benefits, and environmental services that are not accounted for by markets, are also significant. Thus, agriculture supports: environmental services including carbon sequestration, soil conservation, watershed services, biodiversity, and poverty reduction; food security; poverty reduction; agriculture as a social safety net or buffer in times of crisis; and social viability. These indirect contributions are little understood, infrequently examined in the context of development, and infrequently taken into account in the design of national and rural development policies, according to an analysis by the FAO. This might be as a result of the absence of market signals, incorrect policy signals, and a lack of knowledge about the sector's changing market and non-market functions.

In addition to highlighting the numerous cross-sector linkages through which agricultural growth supports overall economic growth and the numerous benefits to society that go beyond what can be evaluated by economic growth indicators alone, the study also eloquently emphasized the close interdependence between agriculture and other sectors. Therefore, based on the extent of this study, a conjecture can be assimilated that the agricultural sector, either directly or indirectly, is indispensably essential for the continuous, sustainable operation of an economy.

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